Preview: y2007_q12 (v1)

Dates: older = born earlier; adding days across month and year ends

Part A: Each question is worth 3 points

1. Basil, who is older than Peter by 1 year and 1 day, was born on August 22, 2013. When was Peter born?

A. August 21, 2012 ***B.*** August 23, 2014 ***C.*** August 21, 2014
D. August 24, 2014 ***E.*** August 23, 2015

2. Omar, who is older than Leo by 1 year and 1 day, was born on January 1, 2005. When was Leo born?

A. January 2, 2006 ***B.*** December 31, 2003 ***C.*** January 1, 2006
D. December 31, 2005 ***E.*** January 3, 2006

3. Sara was born on June 30, 2013. Mia is 1 year and 1 day older than Sara. When was Mia born?

A. June 29, 2011 ***B.*** June 30, 2012 ***C.*** June 29, 2012
D. July 1, 2012 ***E.*** July 1, 2014

4. Leo was born on September 23, 2014. Omar is 1 year and 2 days older than Leo. When was Omar born?

A. September 21, 2012 ***B.*** September 25, 2013 ***C.*** September 23, 2013
D. September 21, 2015 ***E.*** September 21, 2013

Part B: Each question is worth 4 points

5. Omar was born on July 30, 2007 and Leo on August 9, 2007. By how many days is Omar older than Leo?

A. 10 ***B.*** 11 ***C.*** 12 ***D.*** 13 ***E.*** 21

6. Mia was born on October 29, 2005 and Sara on December 6, 2005. By how many days is Mia older than Sara?

A. 47 ***B.*** 45 ***C.*** 42 ***D.*** 39 ***E.*** 38

7. Tom was born on April 9, 2013 and Ben on December 18, 2014. How many years old is Ben on May 15, 2022?

A. 7 ***B.*** 6 ***C.*** 5 ***D.*** 4 ***E.*** 3

8. Mia was born on January 18, 2013 and Sara on November 20, 2014. How many years old is Sara on July 15, 2022?

A. 6 ***B.*** 7 ***C.*** 8 ***D.*** 9 ***E.*** 10

Part C: Each question is worth 5 points

9. Basil and Peter are twins. Basil was born at 11:57 p.m. on December 31, 2012. Peter was born 25 minutes later.
What is Peter's date of birth?

A. December 31, 2013 ***B.*** January 1, 2011 ***C.*** January 1, 2012
D. January 1, 2013 ***E.*** December 31, 2012

10. Tom and Ben are twins. Tom was born at 11:36 p.m. on December 31, 2007. Ben was born 46 minutes later.
What is Ben's date of birth?

A. January 1, 2006 ***B.*** January 2, 2008 ***C.*** December 31, 2008
D. January 1, 2008 ***E.*** January 1, 2007

Answer Key

1. **B**	**2. A**	**3. C**	**4. E**	**5. A**	**6. E**
7. **A**	**8. B**	**9. D**	**10. D**		

Solutions

1. B (August 23, 2014) — Older means born earlier, so Peter was born 1 year and 1 day *after* August 22, 2013: 1 year later is August 22, 2014, then 1 more day: August 23, 2014. [2007 Q12 · younger_birthday]

2. A (January 2, 2006) — Older means born earlier, so Leo was born 1 year and 1 day *after* January 1, 2005: 1 year later is January 1, 2006, then 1 more day: January 2, 2006. [2007 Q12 · younger_birthday]

3. C (June 29, 2012) — Older means born earlier: go back 1 year and 1 day from June 30, 2013. June 30, 2012, then 1 day earlier: June 29, 2012. [2007 Q12 · older_birthday]

4. E (September 21, 2013) — Older means born earlier: go back 1 year and 2 days from September 23, 2014. September 23, 2013, then 2 days earlier: September 21, 2013. [2007 Q12 · older_birthday]

5. A (10) — July has 31 days, so from July 30 to the end of the month is 1 days, then 9 more days to August 9, 2007: 10 days. [2007 Q12 · days_between]

6. E (38) — October has 31 days, so from October 29 to the end of the month is 2 days, then 36 more days to December 6, 2005: 38 days. [2007 Q12 · days_between]

7. A (7) — 2022 – 2014 = 8, but Ben's birthday (December 18) has not come yet that year, so Ben is 7. [2007 Q12 · age_on_day]

8. B (7) — 2022 – 2014 = 8, but Sara's birthday (November 20) has not come yet that year, so Sara is 7. [2007 Q12 · age_on_day]

9. D (January 1, 2013) — 3 minutes after 11:57 p.m. it is midnight, and a new day and a new year start. Peter was born 22 minutes after midnight, on January 1, 2013. So Basil is older, but they have different birthdays and were even born in different years. [2007 Q12 · new_year_twins]

10. D (January 1, 2008) — 24 minutes after 11:36 p.m. it is midnight, and a new day and a new year start. Ben was born 22 minutes after midnight, on January 1, 2008. So Tom is older, but they have different birthdays and were even born in different years. [2007 Q12 · new_year_twins]